

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Built a multi-agent system in Python using the OpenAI Agents SDK, LangGraph, and MCP tool-calling, with prompt engineering and agentic reasoning loops handling millions of daily transactions at 99%+ uptime.
- Designed stateful agent recovery and context management via Redis checkpointing, so interrupted agent workflows resume from the last committed step across distributed, fault-tolerant execution.
- Built and operated AWS/Azure infrastructure (Lambda, SageMaker, Azure AI services) for real-time agentic inference at sub-200ms latency, moving fast from prototype to production alongside product stakeholders.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Improved RAG retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval, semantic re-ranking, and embeddings-based vector search.
- Migrated synchronous PyTorch-based model serving to async vLLM with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests as an early engineer at this AI startup.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Led standardization of failure-handling contracts (retry, timeout, propagation semantics) across 25+ services in partnership with cross-functional teams through code reviews, reducing production escalations 60% in one quarter.

Projects

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with Pinecone vector retrieval and hybrid re-ranking decomposes multi-table questions into dependency-ordered SQL subqueries; deployed as an MCP-compatible server for enterprise data queries.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, SQL, Java, C++

Agentic AI & LLM: Agentic AI, Multi-Agent Systems, LangChain, LangGraph, OpenAI Agents SDK, Anthropic Claude, MCP, RAG, Prompt Engineering, Tool/Function Calling, PyTorch, vLLM, Vector Databases (Pinecone, FAISS)

Backend & Data: FastAPI, Flask, Node.js, REST APIs, Microservices, PostgreSQL, Redis, Distributed Messaging (SQS, Kafka)

Cloud & MLOps: AWS (Lambda, EC2, RDS, SQS, SageMaker), Azure, GCP (Cloud Run, Vertex AI), Docker, Kubernetes, CI/CD, Model Evaluation

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer